Sets

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Based on the lecture note uploaded on SPeCTRUM

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Learning Outcome

At the end of this lecture, you should be able to:

□ Understand what is a set and what is an element of a set.

□ Understand and determine if two sets are equal and if a set is a subset of another.

□ Understand what is a power set of a given set.

□ Understand and perform the set operations.

□ Understand and apply the laws of set algebra.

□ Understand and apply the counting principle to calculate the number of element.

Sets and elements

A set is any well-defined list or collection of objects.

The objects in a set are called elements (members).

We represent a set by listing its elements between curly bracket $\{\}$, separated by commas.

e.g. $A = \{1, 2, 3\}$ is a set with three elements.

Typically, we use an uppercase letter to name a set, while the lowercase letters to name the elements, for example $A = \{a, b, c\}$.

Remark: It is possible to have a set as an element of another set, for example $B = \{1, \{2\}\}$.

Sets and elements

We write $a \in A$ to denote a is an element of the set A, and can be read as "a is in A".

We write $a \notin A$ to denote a is not an element of the set A, and can be read as "a is not in A".

$\{x \mid P(x)\}$ is the set that satisfies the property $P(x)$. The symbol $\mid$ is read as "such that". $\{x \in X \mid P(x)\}$ is read as "the set of all x in X such that $P(x)$".

A set with no element is called the empty (or null) set, and is denoted by $\emptyset$ or $\{\}$.

Example: Which of the following sets are empty?

$A = \{0\}$ $\qquad$ $B = \{x \in \mathbb{R} | x^2 < 0\}$ $\qquad$ $C = \{\emptyset\}$

No
0 is an element of the set.

Yes.
There is no element in B

No.
C does contain an element, $\emptyset$

We say two sets A and B are equal, or $A = B$, if and only if they have the same elements.

Example: Which of the following sets are equal? $(x-2)(x-3)=0$

$A = \{1, 2, 3, 4\}$ $\quad$ $B = \{x \in \mathbb{R} | x^2 - 5x + 6 = 0\}$

$C = \{x \in \mathbb{R} | x \text{ is positive.}\}$ $\quad$ $D = \{x \in \mathbb{Z} | x > 0.\}$

$E = \{x \in A | x \text{ is prime.}\}$ $\quad$ $F = \{x \in \cancel{\mathbb{Z}} | 1 \leq x < 5.\}$

$\mathbb{Z}$

$G = \{x \in \mathbb{R} | 1 \leq x < 5\}$

$B = \{2, 3\}$

$C = \{x \in \mathbb{R}, x > 0\}$

$D = \{1, 2, 3, \ldots\}$

$E = \{2, 3\}$

$F = \{1, 2, 3, 4\}$

$A = F$

$B = E$

Subsets

$A \subseteq B \Rightarrow$ A B or $A = B$

Let A and B be any two sets. We say A is a subset is a subset of B, and write $A \subseteq B$, if and only if every element of A is also an element of B.

If A is a subset of B but $A \neq B$, we say A is a proper subset of B, and write $A \subset B$.

For example: $\{1\}$ is a subset and a proper subset of $\{1, 2\}$.
$\{1, 2\}$ is a subset but not a proper subset of $\{1, 2\}$.

If at least one element of A does not belong to B, we say A is not a subset of B and write $A \not\subset B$ (or $A \not\subseteq B$).

Symbolically, $A \subseteq B$ if and only if $\forall x$, if $x \in A$ then $x \in B$.

A typical format to prove $A \subseteq B$

$$\text{Suppose } x \in A.$$
$$\vdots$$
$$x \in B, \text{ and hence } A \subseteq B.$$

Subsets

The empty set $\emptyset$ is a subset of any set A, that is $\emptyset \subseteq A$.

If A is any set, then $A \subseteq A$.

For any two sets A and B, $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. (We often use this to show two sets are equal)

If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Proof: Given that $A \subseteq B$ and $B \subseteq C$, we want to show that $A \subseteq C$.

Suppose $x \in A$.

Since $A \subseteq B$, $x \in B$.

And since $B \subseteq C$,

$x \in C$ and so $A \subseteq C$.

Distinction between $\in$ and $\subseteq$

Which of the following statements are true?

$16 \in \{x|x \text{ is even.}\}$ T

$\{1,2,3,4,5\}$

$\{4,5\} \subseteq \{x \in \mathbb{N}|x \leq 5\}$ T

$\{1\} \in \{\{1\},\{2\}\}$ T

$\{1\} \in \{1\}$ F, but $\{1\} \in \{\{1\}, 1\}$

$\{\{1\}\} \subseteq \{1,\{2\}\}$ F

& is $\{1\}$ inside $\{1, \{2\}\}$

$\{0\} \subseteq \emptyset$ F, because 0 is not an element of ϕ

$\{1,2\} \in \{1,2,3\}$ F, $\{1,2\} \subseteq \{1,2,3\}$

$\{1,2\} \subseteq \{\{1\},\{2\},\{1,2\}\}$ F, because 1 and 2 are not the element of $\{\{1\},\{2\},\{1,2\}\}$

$\{1,2\} \subseteq \{1,2,\{1,2\}\}$ T, because 1, 2 are elements of $\{1,2,\{1,2\}\}$

$1 \in \{\{1\},\{2\}\}$ F

For $\{a\} \in B$, we need to have the exact thing - $\{a\}$ inside B

For $\{a\} \subseteq B$, we need to have the element in $\{a\}$ which is a inside B.

a number

a set

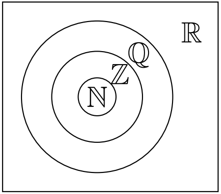
✱ Note: 1 is not equal to $\{1\}$

Sets of numbers

In mathematics, we deals with many different sets of objects; some important sets are the set of all real numbers and its subsets.

Some standard notation used for the sets of numbers:

- $\mathbb{N}$: The set of all natural numbers, $\{1, 2, 3, \dots\}$. * In our course, 0 is NOT a natural number.
- $\mathbb{Z}$: The set of all integers, $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.
- $\mathbb{Q}$: The set of all rational numbers, $\{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\}$.
- $\mathbb{Q}'$: The set of all irrational numbers, $\{x \in \mathbb{R} \mid x \notin \mathbb{Q}\}$.
- $\mathbb{R}$: The set of all real numbers



$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

the set of all positive integers $= \{x \in \mathbb{Z} \mid x > 0\}$

The set of all positive real numbers $= \{x \in \mathbb{R} | x > 0\}$, and
the set of all negative real numbers $= \{x \in \mathbb{R} | x < 0\}$.
Zero is neither positive nor negative.

The set of all non-positive real numbers $= \{x \in \mathbb{R} | x \ngtr 0\}$, and
the set of all non-negative real numbers $= \{x \in \mathbb{R} | x \nless 0\}$.

$\leq$ $\geq$

The set of all odd integers $= \{x \in \mathbb{Z} | x = 2k + 1 \text{ for some } k \in \mathbb{Z}\}$, and
the set of all even integers $= \{x \in \mathbb{Z} | x = 2k \text{ for some } k \in \mathbb{Z}\}$.

Example: Which of the following statements are true? $\in \mathbb{Q}$ $\in \mathbb{Q}$

(i) $-7 \in \mathbb{N}$ F (ii) $\pi \in \mathbb{Q}$ F (iii) $\{0, 0.25\} \subset \mathbb{Q}$ T

(iv) $\sqrt{-3} \in \mathbb{Q}'$ F (v) $\{-\sqrt{4}\} \in \mathbb{Z}$ F (vi) $\{0, \pi, \pi^2\} \subset \mathbb{R}$ T

set

complex number (chapter 3) $\{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$
$\notin \mathbb{R}$

Power set

The set of all subsets of a set A is called the power set of A, and is denoted by $\mathcal{P}(A)$.

For any set A, we have $\emptyset \subseteq A$ and $A \subseteq A$. So we must have $\emptyset \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$.

Example:

1 element

2 elements

4 elements

4 elements

If $A = \emptyset$, the power set of $A = \mathcal{P}(A) = \{\phi\}$

If $B = \{1\}$, the power set of $B = \mathcal{P}(B) = \{\phi, \{1\}\}$

If $C = \{0, 1\}$, the power set of $C = \mathcal{P}(C) = \{\phi, \{0\}, \{1\}, \{0,1\}\}$

If $D = \{\emptyset, \underline{\{1\}}\}$, the power set of $D = \mathcal{P}(D) = \{\phi, \{\phi\}, \{\{1\}\}, \{\phi, \{1\}\}\}$

If $E = \{\underline{1, 2, 3}\}$, the power set of $E = \mathcal{P}(E) = \{\phi, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$

8 elements.

If A has n elements, then $\mathcal{P}(A)$ has 2^n elements.
(A set with n elements has 2^n subsets.)

Extra!

A set A with n elements has 2^n subsets

Proof: Let $B \subseteq A$, for any element x in A, we can have $x \in B$ or $x \notin B$. Since there are n elements in A,

so in total, there are $\underbrace{2 \times 2 \times 2 \times \cdots \times 2}_{\substack{n \text{ times} \\ (n \text{ element})}} = 2^n$ ways to construct the subset B.

This is called counting technique and you will learn this in SIM 2013.

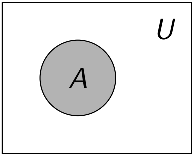
Venn diagram

When working with set, it is often convenient to define a universal set U, which contains all elements of all sets in discussion.

We can use the Venn diagram to visualised the sets as regions in the plane.

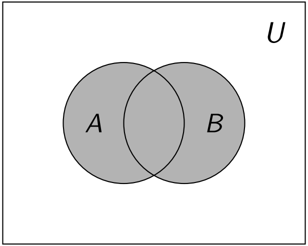
Usually, the universal set U is represented by the interior of a rectangle, and the other sets are represented by disks lying within the rectangle.

Example: Here is a Venn diagram of a universal set U containing a set A.



Union

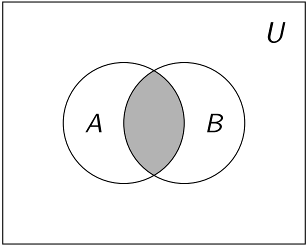
Let A and B be sets and U be the universal set, the union of A and B, denoted by $A \cup B$, is the set $\{x \in U \mid x \in A \text{ or } x \in B\}$.



If $x \in A$ or $x \in B$, then $x \in A \cup B$

Intersection

Let A and B be sets and U be the universal set, the intersection of A and B, denoted by $A \cap B$, is the set $\{x \in U \mid x \in A \text{ and } x \in B\}$.



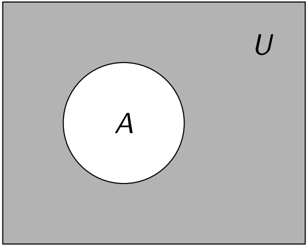
If $x \in A$ and $x \in B$, then $x \in A \cap B$.

If $A \cap B = \emptyset$, then A and B are said to be disjoint or non-intersecting.

Complement

Let A be a set and U be the universal set, the complement of A, denoted by A' or $\overline{A}$, is the set $\{x \in U \mid x \notin A\}$.

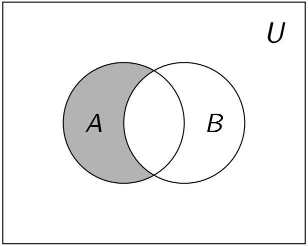
A^c



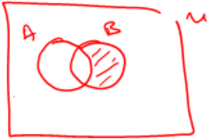
Difference

Let A and B be sets and U be the universal set, the difference of A and B (or relative complement of B in A), denoted by $A - B$, is the set $\{x \in U \mid x \in A \text{ but } x \notin B\}$.

↳ Take the set A, then minus those in B.



B − A



Example (Set operations)

$\{1, 2, 3, 4, 5, \ldots\}$

Let $\mathbb{N}$ be the universal set, $A = \{x \in \mathbb{N} \mid x \leq 6\}$ and $B = \{x \in \mathbb{N} \mid x \text{ is a positive factor of } 12\}$. Find

$A = \{1, 2, 3, 4, 5, 6\}$

$B = \{1, 2, 3, 4, 6, 12\}$

1. $A \cup B = \{1, 2, 3, 4, 5, 6, 12\}$
2. $A \cap B = \{1, 2, 3, 4, 6\}$
3. $A' = \{7, 8, 9, \ldots\} = \{x \in \mathbb{N} \mid x \geq 7\}$
4. $(A \cup B)' = \{7, 8, 9, 10, 11, 13, 14, 15, \ldots\} = \{x \in \mathbb{N} \mid x \geq 7 \text{ and } x \neq 12\}$
5. subset of A: $A - B = \{5\} \neq$ subset of B: $B - A = \{12\}$
6. $(A - B) \cap (A \cup B)' = \phi$

The laws of set algebra

Let A, B, and C be sets of a universal set U, we have the following laws of set algebra:

Idempotent laws	$A \cup A = A$	$A \cap A = A$
Commutative laws	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive laws	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Double complement law	$(A')' = A$	
De Morgan's laws	$(A \cup B)' = A' \cap B'$	$(A \cap B)' = A' \cup B'$
Absorption laws	$A \cup (A \cap B) = A$	$A \cap (A \cup B) = A$
Property of $\emptyset$	$A \cup \emptyset = A$	$A \cap \emptyset = \emptyset$
Property of U	$A \cup U = U$	$A \cap U = A$
Intersection and union with complement	$A \cup A' = U$	$A \cap A' = \emptyset$

The laws of set algebra

Example: Simplify $(A \cup B) \cup (C \cap A) \cup (A \cap B)'$.

$(A \cup B) \cup (C \cap A) \cup (A \cap B)'$

$= (A \cup B) \cup (C \cap A) \cup (A' \cup B')$ De Morgan's Law.

$= (A \cup B) \cup (A' \cup B') \cup (C \cap A)$ Commutative Law.

$= (A \cup A') \cup (B \cup B') \cup (C \cap A)$ Associative and commutative Laws.

$= \mathcal{U} \cup \mathcal{U} \cup (C \cap A)$ Union with complement.

$= \mathcal{U}$ Property of $\mathcal{U}$. and idempotent Law.

The laws of set algebra

Example: Show that $(A \cap U) \cup (B \cap A) = A$.

$(A \cap U) \cup (B \cap A)$

$= A \cup (B \cap A)$ Property of U.

$= A \cup (A \cap B)$ Commutative Law.

$= A$ Absorption Law.

Finite sets

A set is said to be finite if it contains exactly m distinct elements where m is a non-negative integer.

Otherwise, a set is said to be infinite.

If A is a finite set, then the number of elements in A is denoted by $n(A)$ or $|A|$.

Example: Which of the following is a finite set?

$A = \{x \in \mathbb{N} \mid x \leq 5\} = \{1, 2, 3, 4, 5\}$

$|A| = 5$. Finite

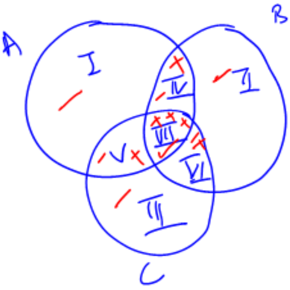
$B = \{x \mid x \text{ is even}\}$

$= \{\ldots, -4, -2, 0, 2, 4, 6, \ldots\}$

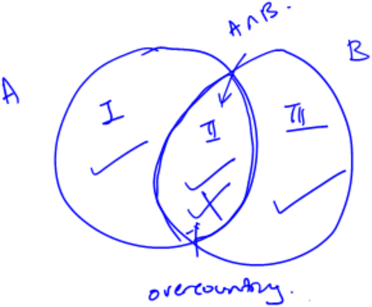
Infinite.

$C = \{x \in \mathbb{R} \mid 1 \leq x \leq 5\}$

Infinite.



$n(A \cup B \cup C)$: no. of elements in I – VII



$n(A \cup B)$: no. of elements in I, II, III

If A and B are finite sets, then $(A \cup B)$ and $(A \cap B)$ are also finite, and $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

If A, B and C are finite sets, then $(A \cup B \cup C)$ and $(A \cap B \cap C)$ are also finite, and $n(A \cup B \cup C) =$
$n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$.

Example: Suppose that 100 of 120 form five students in a secondary school take at least one of the subjects: mathematics, physics and chemistry. Also suppose: 65 take mathematics, 45 take physics, 42 take chemistry and 20 take mathematics and physics, 25 take mathematics and chemistry, and 15 take chemistry and physics.

Determine the number of student who take
(i) all the three subjects,
(ii) only two subjects,
(iii) only one subject,
(iv) none of the subjects.

Counting principle

Let M: the set of all students taking mathematics.

P: the set of all students taking physics.

C: the set of all students taking chemistry.

$\mathcal{U}$: the set of all form five students.

$|\mathcal{U}| = 120$. $\quad |M| = 65 \quad |M \cap P| = 20$

$|M \cup P \cup C| = 100 \quad |P| = 45 \quad |M \cap C| = 25$

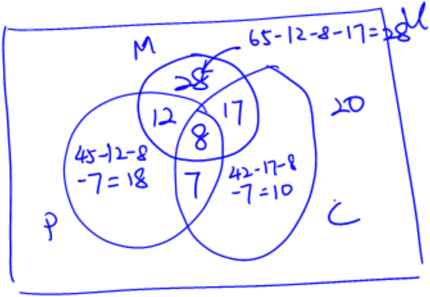
$|C| = 42 \quad |C \cap P| = 15$

(i) The number of student who take all three subjects

$= |M \cap P \cap C|$

$= |M \cup P \cup C| - |M| - |P| - |C| + |M \cap P| + |M \cap C| + |C \cap P|$

$= 100 - 65 - 45 - 42 + 20 + 25 + 15 = 8.$



(ii) From the Venn diagram, the number of student who take only two subjects

$= 12 + 17 + 7 = 36.$

(iii) From the Venn diagram, the number of student who take only one subject

$= 28 + 18 + 10$

$= 56.$

(iv) From the Venn diagram, the number of student who take none of the three subjects

$= 20.$